

Juicetification: Squeeze Control

A hands-on Statistical Process Control simulation — learn SPC by running a juice line

QUICK-START · v1.0



Your job

You're a new quality engineer at Juicetification Inc. Keep the juice in spec: collect a baseline, build the charts, and read the signals.

HOW IT PLAYS · FIVE STAGES

1

Act 1 · Rotterdam

Netherlands

Run 24 shifts on an in-control line and collect a baseline: fill volumes (measured) and defect counts (inspected).



2

Interlude · The Funnel

Netherlands

Adjust vs. leave-alone, shift by shift. See how 'fixing' a stable process (tampering) actually doubles the spread.



3

Act 2 · Blueprints

Netherlands

Pick the right chart, tell sample size n from number of samples k , then compute the centerlines and every control limit yourself.



4

Act 3 · Route 66

United States

Diagnose each week's three charts — catch real signals, ignore noise — and pick an action plan. The plant P&L reacts to every call.



5

Results

Wrap-up

Get your score, detection rate, plant P&L, customer trust, and an LMS completion code to submit.



WHAT YOU'LL LEARN

- X-bar & R for measurements; p-chart for pass/fail
- Control limits from the constants A_2 , D_3 , D_4
- Sample size n vs. number of samples k
- Common cause vs. special cause (the funnel)
- Runs, trends & points beyond the limits
- Type I vs. Type II errors (\$ line stops vs. recalls)

HOW TO RUN IT

1. Install the requirements:

```
pip install -r requirements.txt
```

2. Launch the app:

```
streamlit run juicetification.py
```

3. It opens in your browser (localhost:8501).
4. Use the sidebar to set speed & weeks.
5. Finish → copy your completion code into your LMS.